MATHEMATICS – CLASS X

2016

1. Answer all the questions (1X6=6)
   (i) At a simple rate of interest Rs. 9999 amounts to its double in 10 years. What is the rate of interest?.
   (ii) What is the coefficient of x^0 in the polynomial $2x^3 - 3x^2 + 4x + 5$?
   (iii) What is the HCF of $4p^2qr^3$ and $6p^3q^2r^4$?
   (iv) Find the mixed ratio of $a : bc$, $b : ca$ and $c : ab$.
   (v) ABCD is a cyclic trapezium and $AD \parallel BC$. If $\angle ABC = 70^0$, find the value of $\angle BCD$?
   (vi) Which one of the following is correct? If $\tan\theta = 0$ then:
   (a) $\sin\theta = 0$ (b) $\cos\theta = 0$ (c) $\cot\theta = 0$ (d) None of these.

2. Answer all the questions. (2X7=14)
   (i) Te average of the first 21 natural numbers is 11. What will be the average of first 20 natural numbers?
   (ii) If $xy > 0$, then in which quadrant (x, y) will lie?
   (iii) If x is a perfect square number and $0 < \frac{x-3}{2} < 1$. Find the value of x.
   (iv) In ΔABC, the straight line parallel to the side BC meets AB and AC at the points D and E respectively. If AE = 2AD, find DB : EC.
   (v) The length of the radius of a circle with centre O is 5 cm and the length of the chord AB is 8 cm. What is the distance of the chord AB from the point O?
   (vi) The volume and area of the base of a right pyramid are 40 cubic cm and 24 sq cm respectively. Find its height.
   (vii) From the result $\tan(90^0 - \theta) = \cot\theta,\ 0 < \theta < 90^0$, show that $\cot(90^0 - \theta) = \tan\theta$.

3. Answer any TWO. (5X2=10)
   (a) A man borrowed a sum of money at 5% compound interest per annum. He repays Rs. 3150 at the end of the first year and Rs. 4410 at the end of 2nd year so as to clear the entire loan. How much Rupees did he borrow?
   (b) Milk contains 89% of water. If a sample of milk is found to contain 90% of water, what is the amount of water in 22 litres of such a sample of milk?
   (c) In a mobile phone business of a reputed company, the manufacturer, wholesaler and retailer each of them gains 20%. If a customer purchases a mobile phone of that company for Rs. 8640, find its manufacturing cost.
   (d) The value of a machine in a factory depreciates at the rate of 10% of its value at the beginning of the year. If its value becomes Rs. 43740 after 3 years, what is its present value?

4. Answer any ONE question: (4X1=1)

   (a) Find the HCF of $x^2 - y^2$, $x^3 - y^3$, $3x^2 - 5xy + 2y^2$.

   (b) Find LCM of $ab^4 - 8ab$, $a^2b^4 + 8a^2b$, $ab^4 - 4ab^2$.

5. Solve any ONE (3X1=3)

   (a) $\frac{1}{x} + \frac{5}{y} = \frac{21}{4}$, $\frac{x+y}{x-y} = \frac{5}{3}$.

   (b) $(2x+1)^2 + (x+1)^2 = 4x + 67$.

6. Answer any ONE question: (4X1=4)

   (a) Two numbers are such that one is less than the other by 3 and their product is 70. Find the numbers.

   (b) Length and breadth of a rectangular plot are 5 m more and 3 m less respectively than the length of a square plot. Area of the rectangular plot is less than twice the area of square plot by 78 sq m. Find the length of sides of the square plot.

7. Draw the graphs of the inequalities and indicate the solution region (any ONE). (4X1=4)

   (a) $x + y \le 15$, $x \ge 2$, $y \le -3$.

   (b) $x \ge -7$, $x \le 7$, $y \ge -8$, $y \le 9$.

8. Answer any ONE question: (3X1=3)

   (a) If $\frac{x}{y} = \frac{a+2}{a-2}$ then show that $\frac{x^2 - y^2}{x^2 + y^2} = \frac{4a}{a^2 + 4}$.

   (b) If $bcx = cay = abz$ then show that $\frac{ax + by}{a^2 + b^2} = \frac{by + cz}{b^2 + c^2}$.

9. Answer any ONE question: (3X1=3)

   (a) If $\left(x^3 - \frac{1}{y^3}\right) \propto \left(x^3 + \frac{1}{y^3}\right)$, then prove that $x \propto \frac{1}{y}$.

   (b) Volume of a cone is in joint variation with its square of the radius of base and height. Ratio of radii of base of two cones is 3:4 and ratio of their heights 6:5. Find the raio of their volumes.

10. Answer any ONE question: (3X1=3)

   (a) Simplify $\frac{3\sqrt{7}}{\sqrt{5}+\sqrt{2}} - \frac{5\sqrt{5}}{\sqrt{7}+\sqrt{2}} + \frac{2\sqrt{2}}{\sqrt{5}+\sqrt{7}}$.

   (b) If $x = 2 + \sqrt{3}$, $y = 2 - \sqrt{3}$, then find the value of $\left(xy + \frac{1}{xy}\right)$.

11. Answer any TWO questions: (5X2=10)
   (a) Prove that the angle which an arc of a circle subtends at the centre, is double the angle subtended by it at any point in the remaining part of the circle.
   (b) Prove that a line segment drawn from the centre of a circle to bisect a chord which is not a diameter, is at right angles to the chord.
   (c) Prove that if a perpendicular is drawn from the vertex containing the right angle of a right angled triangle on the hypotenuse the triangles on each side of the perpendicular are similar to each other.

12. Answer any ONE question: (3X1=3)
   (a) There are two concentric circles such that two chords AB and AC of greater circle touch the smaller circle at P and Q respectively. Prove that $PQ = \frac{1}{2}BC$.
   (b) ABCD is a rectangle and O is any point within it. Prove that $OA^2 + OC^2 = OB^2 + OD^2$.

13. Answer any ONE question: (5X1=5)
   (a) Draw a circumcircle of a triangle. (Only traces of construction are required)
   (b) Find Geometrically the value of $\sqrt{21}$. (Only traces of construction are required)

14. Answer any ONE question: (3X1=3)
   (a) A hemisphere and a cone are on equal bases and their heights are also equal. Find the ratio of the area of their curved surfaces.
   (b) Base of a right prism of height 10 cm is a rectangle whose length and breadth are 5 cm and 3 cm respectively. Find the volume of the prism

15. Answer any ONE question: (4X1=4)
   (a) 77 square metre of tarpaulin was needed to make a right circular conical tent. If the slant height of the tent is 7 m, find the area of the base of the tent.
   (b) By melting two solid spheres f radii 1 cm and 6 cm, a hollow sphere of thickness 1 cm is made. Find the outer curved surface area of the new sphere.

16. Answer any TWO questions: (3X2=6)
   (a) A rotating ray revolves in the anti-clockwise direction and makes two complete revolutions from its initial position and moves further to trace an angle of 30^0. What are sexadesimal and circular measure of the angle with reference to trigonometrical measure?
   (b) If α and β are complementary angles to each other, then find the value of $(1-\sin^2\alpha)(1-\cos^2\alpha)(1+\cot^2\beta)(1+\tan^2\beta)$.
   (c) Prove that $\sqrt{\dfrac{1+\cos 30^0}{1-\cos 30^0}} = \sec 60^0 + \tan 60^0$.

(d) If $\cos 52^0 = \frac{x}{\sqrt{x^2 + y^2}}$, thwen find the value of $\tan 38^0$.

17. Answer any ONE question: (5X1=5)

(a) From a quay of a river, 600 metres wide, two boats start in two different directions to reach the opposite side of the river. The first boat moves making an angle 30^0 with this bank and the second boat moves making an angle 90^0 with the direction of the first boat. What will be the distance between the two boats when both of them reach the other side?

(b) The heights of two towers h_1 metres and h_2 metres respectively. If the angle of elevation of the top of first tower from the foot of the second tower is 60^0, and the angle of elevation of the top of the second tower from the foot of first tower is 45^0, then show that $h_1^2 = 3h_2^2$.

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